

PROBLEM SET 4

(Due Feb 10, 2023 @ 8:30 AM in Gradescope and Moodle)

This problem set is to be completed **individually** and **entirely in Excel**. No hardcopies are needed for Problem 1 - Problem 4. You will submit your assignments via Moodle and attach screenshots to the Gradescope. Make sure you start a blank workbook on your own computer. Solve each problem on a separate Spreadsheet; name the Spreadsheets as “Problem 1”, “Problem 2”, and “Problem 3”

Problem 1 (35 points total)

Consider a set of 3 linear equations for 3 unknowns (x, y, and z):

$$2x - 7z = 11$$

$$x + 3y + 2z = -2$$

$$-2y + z = 5$$

- a. (2.5 points) Rewrite the set of equations in the matrix nomenclature:

$$[A] * [X] = [B] \text{ or } A * X = B$$

It will look something like this:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} * \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} b_{11} \\ b_{21} \\ b_{31} \end{bmatrix}$$

- b. (2.5 points) Calculate the determinant of matrix A (*i.e.*, $\det A$) in Excel using the MDETERM function. Note: Follow the steps we outlined in our lectures – watch the lecture again if you need help.
- c. (2.5 points) Determine the inverse matrix of A, *i.e.*, A^{-1} or $[A]^{-1}$ in Excel using the MINVERSE function.
- d. (5 points) Use Excel to calculate the values of [X] (*i.e.*, x, y, and z) by using the matrices you obtained above.
- e. (5 points) Determine the values of x, y, and z by using Solver in Excel.
- f. (5 points) Comment on the answers you have obtained in parts d) and e). Do they match or not?
- g. (7.5 points) Repeat the same steps in a)-e) by considering the following set of 3 linear equations for the same 3 unknowns (*i.e.*, x, y, and z):

$$2x + 7z = 11$$

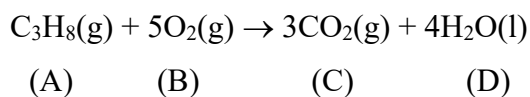
$$x + 3y + 2z = -2$$

$$-2y + z = 5$$

- h. (5 points) Comment on the answers you have obtained in part g). Do both methods generate similar results or not? Why?

Problem 2 (35 points total)

Propane is combusted with oxygen in a reactor following the reaction equation below:



The input into the reactor contains 100 mol/s of propane, 10 mol/s of carbon dioxide, and an unknown amount of oxygen. The output contains 200 mol/s of O₂ and unknown amounts of the remaining components. No propane is found in the output stream. Use the following nomenclature to answer questions below: 1) use A for C₃H₈, B for O₂, C for CO₂, and D for H₂O; 2) use $\dot{n}_{x0}$ (e.g., $\dot{n}_{A0}$, $\dot{n}_{B0}$, etc.) to designate the flow of x species into the reactor; 3) use $\dot{n}_{x1}$ (e.g., $\dot{n}_{A1}$, $\dot{n}_{B1}$, etc.) to designate the flow of x species from the reactor. In this problem you do not need to draw a flow chart. You do not need to perform the DOF analysis. However, we still recommend you do it for yourself.

- (10 points) Using the **extent of reaction (EOR)** method, write down all balance equations needed to solve the unknowns. Write these balance equations directly into the spreadsheet.
- (7.5 points) Rearrange the balance equations into matrix formalism so that they look like $[A]*[X]=[B]$, where $[X]$ is a matrix of your unknown variables. Specify each matrix (e.g., $[A]$, $[X]$, and $[B]$) in the spreadsheet.
- (2.5 points) Calculate the determinant of matrix A. Note: you can use excel's default function.
- (5 points) Determine the unknowns in matrix $[X]$ by using $[X]=[A]^{-1}* [B]$. Note: you can use excel's default function.
- (10 points) Determine the values of the unknowns using Solver.

Problem 3 (30 points total)

Use Solver in Excel to find ALL possible solutions to the following set of equations, in which a, b, c, and d are variables that belong to the group of real numbers:

$$-2a + 5b - 2c^2 = 11$$

$$a + 5b - 8d = -2$$

$$3a + 8c^2 - 3d = 3$$

$$3c^2 - 6d = 9$$